MATH I – Week 7

Practical Session exercises

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Exercise 1

"differentiate" means "find the first derivative"

Differentiate:

a. $3\sin x - 2\cos x.$

b. $\tan x + \cot x.$

c. $\sec x - \csc x.$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x.$$

Exercise 1

"differentiate" means "find the first derivative"

Solution

Differentiate:

a. $3 \sin x - 2 \cos x$.

b. $\tan x + \cot x$.

c. $\sec x - \csc x$.

Solution

a. $\boxed{3 \cos x + 2 \sin x}$.

b. $\boxed{\sec^2 x - \csc^2 x}$.

c. $\boxed{\sec x \tan x + \csc x \cot x}$.

Exercise 2

basic formulas are shown

Derive the derivative of $\tan x$

Starting from $\tan x = \sin x / \cos x$, use the quotient rule to derive its derivative.

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

Theorem 3.6

The Quotient Rule

Let $f(x)$ and $g(x)$ be differentiable functions. Then

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{\frac{d}{dx}(f(x)) \cdot g(x) - \frac{d}{dx}(g(x)) \cdot f(x)}{(g(x))^2}.$$

That is,

$$\text{if } j(x) = \frac{f(x)}{g(x)}, \text{ then } j'(x) = \frac{f'(x)\,g(x) - g'(x)\,f(x)}{(g(x))^2}.$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x.$$

Exercise 2
Solution

Derive the derivative of $\tan x$

Starting from $\tan x = \sin x / \cos x$, use the quotient rule to derive its derivative.

Solution

$$\frac{d}{dx}\tan x = \frac{\cos x \cos x - \sin x(-\sin x)}{\cos^2 x}.$$

Because $\cos^2 x + \sin^2 x = 1$,

$$\boxed{\frac{d}{dx}\tan x = \frac{1}{\cos^2 x} = \sec^2 x}, \qquad \cos x \neq 0.$$

Exercise 3

Differentiate

$$f(x) = \frac{\sin x}{1+\cos x}, \qquad 1 + \cos x \neq 0.$$

Exercise 3
Solution

Differentiate

$$f(x) = \frac{\sin x}{1+\cos x}, \qquad 1 + \cos x \neq 0.$$

Solution

$$f'(x) = \frac{\cos x(1+\cos x)+\sin^2 x}{(1+\cos x)^2}.$$

The numerator simplifies to $1 + \cos x$, giving

$$\boxed{f'(x) = \frac{1}{1 + \cos x}},$$

on the original domain.

Exercise 4

A vibrating component has displacement

$$s(t) = 0.02 \cos t \ \mathrm{m},$$

where the angular frequency is $1 \ \mathrm{rad/s}$.

Find velocity and acceleration. Evaluate them at $t = \pi/2$ seconds. Also find $d^4 s/dt^4$.

Exercise 4 Solution

A vibrating component has displacement

$$s(t) = 0.02\cos t \text{ m},$$

where the angular frequency is 1 rad/s.

Find velocity and acceleration. Evaluate them at $t = \pi/2$ seconds. Also find d^4s/dt^4.

Solution

$$\boxed{v(t) = -0.02\sin t \text{ m/s}}, \qquad \boxed{a(t) = -0.02\cos t \text{ m/s}^2}.$$

At $t = \pi/2$,

$$\boxed{v = -0.02 \text{ m/s}, \qquad a = 0}.$$

Differentiating twice more,

$$s'''(t) = 0.02\sin t, \qquad \boxed{s^{(4)}(t) = 0.02\cos t \text{ m/s}^4}.$$

The fourth derivative repeats the original trigonometric form.

Exercise 5

Beware of the different bases

Differentiate

$$f(x) = 3e^x + 2^x - 5.$$

Then find $f'(0)$.

For

$$f(x) = 2^x,$$

the derivative is

$$\boxed{f'(x) = 2^x \ln 2.}$$

Step 1: Rewrite using the natural exponential

Since $2 = e^{\ln 2}$,

$$2^x = \left(e^{\ln 2}\right)^x = e^{x \ln 2}.$$

Thus,

$$f(x) = e^{x \ln 2}.$$

Exercise 5

Beware of the different bases

Solution

Differentiate

$$f(x) = 3e^x + 2^x - 5.$$

Then find $f'(0)$.

Solution

Use $(e^x)' = e^x$ and $(a^x)' = a^x \ln a$:

$$\boxed{f'(x) = 3e^x + 2^x \ln 2}.$$

Therefore,

$$\boxed{f'(0) = 3 + \ln 2}.$$

Differentiate

Exercise 6

$$f(x) = 2\ln x - \log_{10} x, \qquad x > 0.$$

Find the tangent line at $x = 1$.

The **natural logarithm**, $\ln x$, has base e. Its derivative is

$$\boxed{\frac{d}{dx}\ln x = \frac{1}{x}}, \qquad x > 0.$$

The **base-10 logarithm**, $\log_{10} x$, has derivative

$$\boxed{\frac{d}{dx}\log_{10} x = \frac{1}{x\ln 10}}, \qquad x > 0.$$

To differentiate a logarithm with another base, use the **change-of-base formula**:

$$\log_a x = \tfrac{\ln x}{\ln a}, \qquad a > 0, \quad a \neq 1.$$

Since $\ln a$ is constant,

$$\boxed{\frac{d}{dx}\log_a x = \frac{1}{x\ln a}}.$$

Exercise 6
Solution

Differentiate

$$f(x) = 2\ln x - \log_{10} x, \qquad x > 0.$$

Find the tangent line at $x = 1$.

Solution

$$\boxed{f'(x) = \frac{2}{x} - \frac{1}{x \ln 10}}.$$

Since $f(1) = 0$, the tangent line is

$$\boxed{y = \left(2 - \frac{1}{\ln 10}\right)(x - 1)}.$$

Exercise 7

Differentiate:

a. $f(x) = x \ln x,\ x > 0.$

b. $g(x) = e^x / x,\ x \neq 0.$

Exercise 7
Solution

Differentiate:

a. $f(x) = x \ln x$, $x > 0$.

b. $g(x) = e^x / x$, $x \neq 0$.

Solution

a. Product rule:

$$f'(x) = 1 \cdot \ln x + x \cdot \tfrac{1}{x} = \boxed{\ln x + 1}.$$

b. Quotient rule:

$$g'(x) = \tfrac{xe^x - e^x}{x^2} = \boxed{\frac{e^x(x-1)}{x^2}}.$$

Exercise 8

Find the first derivative of the function $f(x) = x^2 e^x$

Exercise 8
Solution

Find the first derivative of the function $f(x) = x^2 e^x$

Solution: use the product rule

The function is a product of $u(x) = x^2$ and $v(x) = e^x$. The product rule states

$$(uv)' = u'v + uv'.$$

Since

$$u'(x) = 2x, \qquad v'(x) = e^x,$$

we obtain

$$f'(x) = 2xe^x + x^2 e^x.$$

Factoring gives

$$\boxed{f'(x) = e^x(x^2 + 2x) = xe^x(x + 2), \qquad x \in \mathbb{R}.}$$

Exercise 9

Exponential decay in an electrical circuit

A discharging capacitor has voltage

$$V(t) = 12e^{-t/2}\ \text{V}, \qquad t \geq 0,$$

with time in seconds.

Find $V'(t)$, the initial rate of voltage change, and $V'(2)$.

Exercise 9 Solution

Exponential decay in an electrical circuit

A discharging capacitor has voltage

$$V(t) = 12e^{-t/2}\ \text{V}, \qquad t \geq 0,$$

with time in seconds.

Find $V'(t)$, the initial rate of voltage change, and $V'(2)$.

Solution

The derivative of the exponent $-t/2$ is $-1/2$. Therefore,

$$V'(t) = 12e^{-t/2}\left(-\tfrac{1}{2}\right) = \boxed{-6e^{-t/2}\ \text{V/s}}.$$

Initially,

$$\boxed{V'(0) = -6\ \text{V/s}}.$$

At $t = 2$,

$$\boxed{V'(2) = -\frac{6}{e}\ \text{V/s} \approx -2.21\ \text{V/s}}.$$

The negative sign means voltage is decreasing; its rate of decrease becomes smaller in magnitude over time.

REVIEW
Exercise 10

Charge and electric current

The charge passing a point in a wire is

$$q(t) = t^3 + 2t^2 + 3t$$

Coulombs. Find the electric current at $t = 2$, the average current over $[0, 2]$, and the rate of change of current at $t = 2$.

Note: the electric current is $I(t) = q'(t)$

REVIEW
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Solution

Charge and electric current

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Note: the electric current is $I(t) = q'(t)$

Solution

$$I(t) = q'(t) = 3t^2 + 4t + 3.$$

Therefore,

$$\boxed{I(2) = 23 \text{ A}}.$$

Since $q(2) = 22$ and $q(0) = 0$,

$$\boxed{I_{\text{avg}} = \frac{22}{2} = 11 \text{ A}}.$$

Finally,

$$I'(t) = 6t + 4, \qquad \boxed{I'(2) = 16 \text{ A/s}}.$$

REVIEW
Exercise 11

Electrical power

For $0 \leq t \leq 4$, a device has voltage $V(t) = 12 - t$ volts and current $I(t) = 1 + t$ amperes.

Find the electric power $P(t)$, $P'(t)$, and their values at $t = 2$.

Note: the electric power is $P(t) = V(t)I(t)$.

REVIEW
Exercise 11
Solution

Electrical power

For $0 \le t \le 4$, a device has voltage $V(t) = 12 - t$ volts and current $I(t) = 1 + t$ amperes.

Find the electric power $P(t)$, $P'(t)$, and their values at $t = 2$.

Note: the electric power is $P(t) = V(t)I(t)$.

Solution

$$P(t) = V(t)I(t) = (12 - t)(1 + t) = \boxed{12 + 11t - t^2 \text{ W}}.$$

Using the product rule,

$$P' = V'I + VI' = -(1 + t) + (12 - t) = \boxed{11 - 2t \text{ W/s}}.$$

Thus,

$$\boxed{P(2) = 30 \text{ W}, \qquad P'(2) = 7 \text{ W/s}}.$$

At that instant, the power consumption is increasing.